Aayushi Sinha

sinhaaayushi19@gmail.com | +91 9335709472 | linkedin.com/in/aayushi-sinha-43b0a0281 | github.com/Sinna19

AI engineer who builds LLM agents and the infrastructure that makes them safe to run: RAG and tool-calling agents, an MCP tool gateway deployed on Kubernetes, and evaluation suites with calibrated LLM judges that catch regressions before release. B.Tech IT graduate (2026) with shipped, tested, documented projects on GitHub.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

LLM Agents & RAG: Tool / function calling, Model Context Protocol (MCP), Retrieval-Augmented Generation, hybrid search (TF-IDF + sentence-transformers, reciprocal-rank fusion), Groq, OpenAI-compatible APIs, prompt-injection defenses

Evaluation & Testing: LLM-as-judge with judge calibration, multi-turn conversation evals, regression gates, retrieval benchmarks (Recall@k, MRR, Hit@k), pytest, Vitest

Backend & Infrastructure: Node.js, Express, React, PostgreSQL, MySQL, Flask, Docker, Kubernetes (k3s), Helm, cert-manager, GitHub Actions

Security & Observability: Role-based access control, SSRF protection, NetworkPolicies, AES-256-GCM secret encryption, rate limiting, audit logging, structured JSON tracing

Tools: Git, GitHub, scikit-learn, Streamlit, Claude Code (AI-assisted development)

PROJECTS

MCP Gateway on Kubernetes | GitHub Oct 2026

TypeScript, Node.js, MCP SDK, Express, PostgreSQL, React, Kubernetes (k3s), Helm, cert-manager, Vitest

- Built a team MCP gateway that aggregates 3 upstream MCP servers (WooCommerce store, FX rates, notes) behind one authenticated Streamable HTTP endpoint, with namespaced tools, live `tools/list_changed` updates, and per-role tool permissions enforced on listing and on every call.
- Added an LLM chat agent that calls tools through the same auth, role and audit path as external MCP clients, so a member cannot do more in chat than with an API key; write actions require explicit user confirmation.
- Hardened the platform with an SSRF guard backed by egress NetworkPolicies, default-deny ingress, non-root containers, AES-256-GCM-encrypted upstream credentials with key rotation, hashed API keys revocable mid-session, rate limits, and a per-user audit log.
- Deployed with Helm to k3s on an Azure VM with Let's Encrypt TLS, a private API server, a rehearsed `helm rollback`, and nightly Postgres backups with automated restore checks; 9/9 end-to-end checks pass over HTTPS, CI runs lint and tests on every push, and the design is recorded in 25 ADRs plus a threat model.

Aster & Row Support Agent | GitHub Sep 2026

Python, Groq API (tool calling), sentence-transformers, scikit-learn, pytest, Streamlit

- Built and deployed a RAG + tool-calling customer support agent with hybrid retrieval (TF-IDF fused with BGE embeddings via reciprocal-rank fusion), raising Recall@3 from 83% to 100% and MRR from 0.75 to 0.90 over lexical-only search on a labelled benchmark.
- Added a source-authority layer that prefers active policy over superseded or draft documents, and a flagged non-authoritative lane so the agent recognises and dismisses injected content instead of following it.
- Built a privacy-safe order-lookup tool with a strict field allow-list so PII never enters the model's context, and traced every turn (retrieval scores, tool calls, handoff decisions) as structured JSON.
- Found and root-caused 9 bugs through a 23-case evaluation suite (retrieval-authority conflicts, invisible Unicode breaking exact-match evals, order-ID parsing) and added a regression test for each; 34/34 unit tests pass in CI.

AI Eval Harness | GitHub Aug 2026 – Present

Python, Groq API, LLM-as-judge, REST APIs, matplotlib

- Built a harness that grades whole conversations for a policy-bound support agent: 24 cases, including 8 multi-turn scripted customers who lie, escalate and rephrase, scored on consistency, policy adherence, pressure resistance and memory against a 22-rule policy.
- Combined deterministic keyword checks, regex gates and an LLM judge from a different model than the agent; calibrated the judge on 16 hand-labelled transcripts (15/16 correct, each catch naming the violated turn and rule).
- Added a baseline regression gate, a repeat-run stability check and 62 offline unit tests; separating API errors from agent failures cut 11 reported "flaky" cases to 3 real ones, and the latest run passes 24/24.

EDUCATION

B.Tech, Information Technology — Chandigarh University | CGPA: 7.57 2022 – 2026

Class XII (ISC) — City Montessori School, Lucknow | 95.5% 2020 – 2021

Class X (ICSE) — City Montessori School, Lucknow | 88.5% 2018 – 2019

ACHIEVEMENTS

- **Top 100, UnvibeCode Engineering Challenge 2026** — Alphashots.ai, Sept 2026 | ID: UVC26-TOP100-054 | Verify